IoT

Article

Adaptive Federated Baseline K-Means for Lightweight IoT Intrusion Detection: Auto-Thresholding and Robust Statistics Aggregation

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Abstract

Federated, semi-supervised novelty detection is well suited for intrusion detection on resource-constrained Internet of Things (IoT) nodes: each device learns a model of benign traffic, shares only summary statistics, and does not transmit raw traffic samples. A previously published cross-layer federated detector, *Baseline K-Means*, showed that periodically merging worker statistics through a coordinator raises the detection rate, but it also exhibited a systematic side effect: after every merge, the precision decays, and the false-positive rate (FPR) climbs because the coordinator recomputes its threshold from streaming distances filtered by the closest observed anomaly, so tightens after every merge, flagging progressively more benign traffic; the threshold was also hand-tuned. We present AF-BKM, an Adaptive Federated Baseline K-Means that repairs the federated mechanism with two label-free, statistics-only enhancements, denoted as E1 and E2: (i) an adaptive decision threshold read from the benign Mahalanobis-distance distribution, requiring no manual percentile search and no attack labels (E1), and (ii) a robust, benignly anchored aggregation that blends worker means under quality weighting and outlier-worker filtering and recalibrates the threshold on a trusted benign anchor to a stable, anchor-referenced false-positive level, which a target-FPR rule can make operator-selectable instead of tightening it toward the nearest anomaly (E2). With MinMax scaling fit only on benign baseline data and non-IID federated streams on NSL-KDD, UNSW-NB15 and the N-BaIoT corpus of real traffic from commercial IoT devices, AF-BKM removes the merge-induced precision decay (the first-to-last-epoch precision change improves from -0.134 to -0.002 on NSL-KDD, from -0.121 to -0.014 on UNSW-NB15, and from -0.170 to -0.009 on N-BaIoT) and reduces the mean FPR by 30–64%, depending on the dataset; all central improvements are significant across 10 seeds (Wilcoxon $p = 0.002$, large effect sizes). AF-BKM preserves recall on NSL-KDD and N-BaIoT and, on the harder UNSW-NB15, exposes an explicit precision–recall trade-off through a benign target-FPR knob. In fp32, the deployed model serializes to 5.5–52 KB, a packet is classified in 11–27 µs on a desktop CPU, and each merge round uploads a d+3-value summary (160–472 B) 94.6–98.3% smaller than the same summary extended with the covariance upper triangle. A robustness study covering selected faulty-worker updates, contamination of the commissioning anchor, and detector-level white-box evasion reports the measured degradation patterns: fabricated threshold candidates have no direct path to the threshold, although a fabricated mean still reaches it indirectly through the blended centroid, and the anchor-referenced false-positive level remains stable under

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Academic Editors: Jingyi He, Jiabao Wen and Meng Xi

Received: 9 June 2026
Revised: 12 August 2026
Accepted: 14 August 2026
Published: 21 August 2026

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percent-level anchor contamination, while recall sensitivity is dataset-dependent and the evasion budget tracks the benign–attack margin of each dataset. We frame the contribution with a focused taxonomy that identifies merge-induced precision decay under non-IID workers as an open gap. Code is released for reproducibility.

Keywords: intrusion detection; Internet of Things; federated learning; semi-supervised novelty detection; Mahalanobis distance; lightweight IDS; non-IID; adaptive thresholding

1. Introduction

The proliferation of Internet of Things (IoT) devices has enlarged the attack surface of modern networks while concentrating it on nodes constrained in memory, computation and energy [1,2]. An intrusion detection system (IDS) for such nodes has to satisfy three requirements at once: it must be light enough to run on the device, it must not expose the device's traffic, and it should still benefit from what other devices observe. How the learning is organized determines how well these requirements can be met. The classical option is centralized training, in which every node ships its traffic to a server that fits one global model; this concentrates sensitive traffic in a single place, consumes uplink bandwidth that constrained deployments rarely have, and creates a single point of failure [2]. The opposite option is fully local training with one isolated model per device; it avoids data movement, but each node then learns from a small and biased sample of the threat landscape and never profits from attacks first seen elsewhere. Federated learning (FL) is the middle ground between these two distributed extremes: nodes train locally and exchange only model parameters or statistics, so collaboration survives without raw-data transfer [3–5]. Complementary work secures the IoT data path itself through lightweight cryptographic and data-hiding primitives suited to constrained nodes [6]. A separate constraint concerns labels: attack labels are scarce on deployed devices, whereas benign traffic is abundant, which makes semi-supervised *novelty* detection, trained on abundant benign traffic rather than on labeled attacks, the practical choice [7].

This paper follows a lightweight IoT IDS research line that first reduces the traffic processed by the detector through cluster-based sampling [8], then introduces the cross-layer federated *Baseline K-Means* detector [9]. The latter adapts k-means by replacing the Euclidean metric with the Mahalanobis distance to a benign centroid and flags packets whose distance exceeds a percentile threshold. Each worker shares only means and distances; a coordinator periodically merges these and redistributes a global model. That work reports a useful property (merging raises recall over time) but also a recurring weakness: after each merge, the *precision* falls, and the FPR rises [9]. The cause is structural. The coordinator recomputes the threshold as a percentile of benign distances *filtered to those below the nearest worker anomaly*; in the evaluated windowed protocol, this rule repeatedly tightens the threshold and rejects ever more benign packets. A second, practical weakness is that the percentile was "set based on experience, including trial-and-error experimentation" [9].

A natural reaction is to add an ensemble of auxiliary detectors to suppress false positives. This is the direction of the subsequent federated ensemble extension of Baseline K-Means, where local novelty detectors are combined around BKM to reduce false alerts after the base detector has already produced a decision [10]. That extension is valuable, but it acts *downstream*: it does not address *why* the coordinator becomes miscalibrated after a merge. AF-BKM instead repairs the federated statistics/threshold mechanism itself, which is the root cause of the decay. AF-BKM optimizes *alert stability and false-positive control* rather than maximal recall on every dataset: on NSL-KDD, it preserves recall,

whereas on the harder UNSW-NB15, it trades recall for precision along an operator-chosen target-FPR curve. Figure 1 places these four steps in sequence.

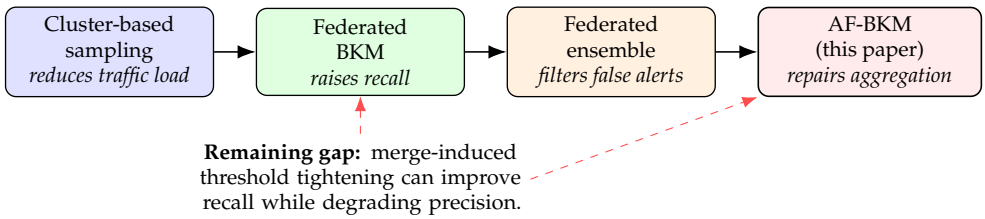


Figure 1. Evolution of the lightweight BKM IoT IDS line, from cluster-based sampling [8] through the cross-layer federated detector [9] and its ensemble extension [10] to AF-BKM. The published cross-layer federated system improves collaborative recall but exposes merge-induced precision decay; the ensemble extension reduces false positives downstream; AF-BKM complements both by repairing the coordinator aggregation and thresholding mechanism.

We propose **AF-BKM** (Adaptive Federated Baseline K-Means), which keeps the lightweight, statistics-only, semi-supervised identity of Baseline K-Means and repairs its federated mechanism. The contributions are three framework improvements:

1. **Diagnosis and formalization of merge-induced precision decay.** We show that the decay is a structural property of the original coordinator rule, whose threshold tightens toward the nearest worker-reported anomaly, and that label-skew heterogeneity amplifies it (Section 2); a focused taxonomy of lightweight, federated, semi-supervised IoT IDS places this failure mode in the literature and shows it is a sparsely addressed gap (Section 3).
2. **E1: adaptive auto-thresholding.** The decision threshold is read from the shape of the benign Mahalanobis-distance distribution, through a robust median-absolute-deviation rule or a target-FPR calibration, which removes the manually searched percentile; each worker also reports its scalar threshold candidate, which the coordinator uses to filter outlier workers (Section 4).
3. **E2: robust, benignly anchored aggregation.** The coordinator blends worker means with a trusted benign anchor under count and quality weighting behind a trust gate and recalibrates the threshold on that anchor to a stable, anchor-referenced false-positive level, with an operator-selectable target-FPR variant, replacing the nearest-anomaly tightening that causes the decay; both enhancements are label-free with respect to attack labels and preserve the statistics-only communication model (Section 4).

We validate AF-BKM on NSL-KDD, UNSW-NB15 and the real-device N-BaIoT traffic corpus under a non-IID federated protocol with baseline-only MinMax scaling and disclosed label-blind feature filtering, with a four-way ablation, significance testing, heterogeneity and scale sweeps, adversarial stress tests (selected faulty-worker updates, anchor contamination and white-box evasion) and a computational resource profile; the implementation and all experimental scripts are publicly available for reproducibility.

The remainder of this paper is organized as follows. Section 2 defines the problem: it summarizes the original BKM update rule and the failure mode that motivates AF-BKM. Section 3 reviews lightweight, federated, and ensemble IDS work and positions the gap. Section 4 presents the proposed aggregation and thresholding method. Sections 5 and 6 describe the experimental protocol and results. Section 7 discusses limitations and future work, and Section 8 concludes the paper.

https://doi.org/10.3390/iot7030067

2. Problem Definition

Let $X = \{x_1, \ldots, x_n\}$ and $x_i \in \mathbb{R}^d$ be benign baseline observations with mean (μ) and covariance (Σ). The Mahalanobis distance to the benign centroid is

$$D_M(x) = \sqrt{(x-\mu)^\top \Sigma_\lambda^{-1}(x-\mu)}, \qquad \Sigma_\lambda = \Sigma + \lambda I, \tag{1}$$

with a small ridge (λI) for numerical stability. A threshold (τ) is a percentile of the benign distances, and x' is flagged as anomalous iff $D_M(x') > \tau$. Only benign data are needed, and inference is one quadratic form, which is cheap enough to run on resource-constrained IoT edge or gateway nodes.

In the federated version, W workers receive $(\mu, \Sigma_\lambda^{-1}, \tau)$ at initialization, classify their local window, and upload only means and distances. The original coordinator-side merge averages means and recomputes the threshold [9]:

$$\mu \leftarrow \tfrac{1}{2}(\mu_{\text{coord}} + \mu_{\text{worker}}), \tag{2}$$

$$d_{\min} = \min_w \min_i D_M\big(a_i^{(w)}\big), \tag{3}$$

$$\tau \leftarrow \text{percentile}\big(\{D_M(x) : x \in B,\ D_M(x) < d_{\min}\},\ p\big), \tag{4}$$

where $a_i^{(w)}$ are points classified as anomalous by worker w and B is the pooled benign set. Equation (4) is the culprit: each merge admits only the benign distances that fall below the closest anomaly reported in that round, and in the evaluated windowed protocol, this min-outline filtering repeatedly tightens the threshold. The consequence quantified in Section 6, is rising recall but falling precision and climbing FPR after each merge. The effect is amplified under non-IID workers: $d_{\min}$ in Equation (3) is the minimum over all workers in the round, so a single worker whose local stream contains a near-boundary anomaly lowers the shared threshold for everyone, and the heterogeneity sweeps in Section 6 show the decay across the full non-IID range. Moreover, the percentile (p) is chosen manually.

The problem this paper addresses is therefore twofold: replace the manual percentile with a threshold rule that requires no per-dataset percentile search, though it is not parameter-free, and replace the merge rule of Equations (2)–(4) with an aggregation whose false-positive behavior stays calibrated across merges, without giving up the statistics-only communication model.

3. Related Work and Taxonomy

Figure 2 organizes the design space along six axes: *learning paradigm*, *federation* style, *deployment* target, *detection model*, *data dynamics*, and *threshold/robustness*. Table 1 lists representative works per axis.

Table 1. Taxonomy axes for lightweight, federated, semi-supervised IoT IDS with representative works from the surveyed corpus. The last axis is sparsely populated and motivates this work.

Axis	Categories	Representative Works
Learning paradigm	Supervised; Unsupervised; Semi-supervised novelty / one-class	Belarbi et al. 2023 [11]; Saez-de-Camara et al. 2023 [12]; Blanchard et al. 2010 [7], Shrivastava and Vamsi 2026 [13], Nguyen and Beuran 2025 [14]

Table 1. *Cont.*

Axis	Categories	Representative Works
Federation	Centralized; FL (weights); FL (statistics-only); Decentralized	Shrivastava and Vamsi 2026 [13]; Belarbi et al. 2023 [11], Attota et al. 2021 [15]; Bouzinis et al. 2025 [16], Hajj et al. 2023 [9]; Nakip and Gelenbe 2023 [17]
Deployment	Cloud; Fog/edge gateway; Resource-constrained IoT edge	Campos et al. 2022 [1]; Attota et al. 2021 [15]; Hajj et al. 2023 [9], this work
Detection model	Deep learning; Tree/ensemble; Clustering + Mahalanobis; One-class	Belarbi et al. 2023 [11]; Attota et al. 2021 [15]; Lima et al. 2010 [18], Melnykov and Melnykov 2014 [19], Hajj et al. 2023 [9]; Blanchard et al. 2010 [7]
Data dynamics	Batch; Streaming/sliding window; Concept-drift/continual	(most works); Wu et al. 2025 [20], Hajj et al. 2023 [9]; Gama et al. 2014 [21], Nakip and Gelenbe 2023 [17]
Threshold and robustness	Fixed/manual; Adaptive; Robust aggregation (anti-decay)	Hajj et al. 2023 [9] (manual); this work

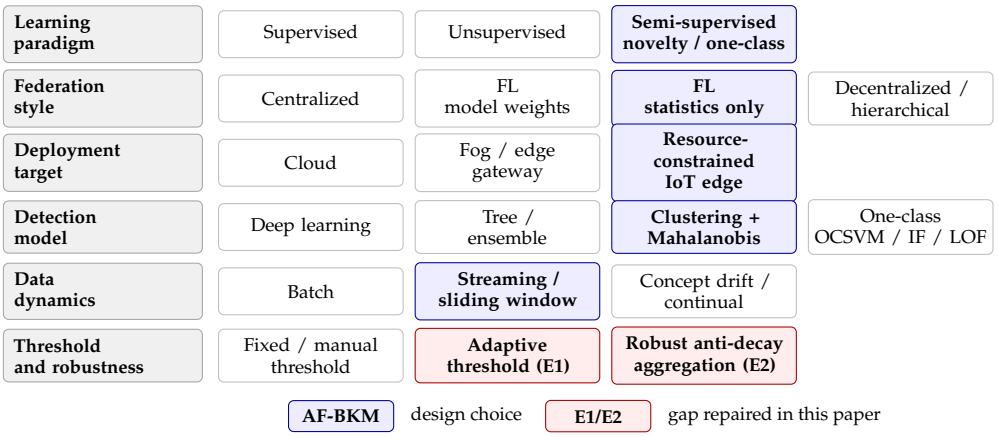


Figure 2. Taxonomy of lightweight, federated, semi-supervised IoT intrusion detection. Blue cells mark the AF-BKM design path; red cells mark the specific thresholding and aggregation gap repaired by this paper.

Sharing *statistics* rather than weights is a small but growing branch of federated IDS: StatAvg [16] shares per-feature means and variances for non-IID normalization, and Baseline K-Means [9] shares centroid means and distances. FedMSE [14] is the closest semi-supervised federated novelty detector but learns a shrink-autoencoder and aggregates weights heavier than a Mahalanobis centroid. Within the BKM line, the federated ensemble extension [10] reduces false positives by surrounding BKM with auxiliary local novelty detectors; AF-BKM instead keeps the detector single-stage and targets the coordinator threshold/merge rule that creates the false positives. Mahalanobis-based one-class detection for IoT was recently revisited by Shrivastava and Vamsi [13] but centralized on host/sensor data; the benign-baseline-plus-*k*-means idea dates to Lima et al. [18], and *K*-means with Mahalanobis distances dates to Melnykov and Melnykov [19], neither federated nor lightweight. Decentralized online FL [17] and online ensembles [20] address streaming and drift but rely on deeper or ensemble models and do not target the precision-

decay phenomenon; federated ensembles [15] and clustered FL [12] improve heterogeneity handling but are supervised or unsupervised. Unsupervised deep-learning detectors have also been applied to streaming smart-grid communication traffic [22] but rely on heavier representations than a Mahalanobis centroid. Tree-based and tiered detectors offer another lightweight centralized alternative for IoT intrusion detection [23], though they remain non-federated. Surveys confirm FL for IoT IDS is well studied [1,2,4,5]. Table 2 summarizes the comparison: among the works compared there, AF-BKM is the only one that explicitly targets merge-induced precision decay while remaining statistics-only, semi-supervised, and aligned with resource-constrained IoT edge/gateway settings.

Table 2. Differential analysis against the closest IoT/network IDS works. * Ensemble extension exists in prior work but is orthogonal here. †AF-BKM adapts its *threshold* to the benign distance distribution but implements no streaming concept-drift mechanism; drift adaptation is future work (Section 7). Among the compared works, AF-BKM is the only one that explicitly targets the merge-induced precision decay (Anti-decay) while remaining statistics-only, semi-supervised, and aligned with resource-constrained IoT edge/gateway settings.

Work	Paradigm	Federation/Shared	IoT Setting	Drift	Ensemble	Datasets	Anti-Decay
Hajj 2023 [9] (prior)	Semi-sup novelty	FL/statistics	Federated IoT IDS	No	No *	NSL-KDD	No
FedMSE 2025 [14]	Semi-sup (SAE-CEN)	FL/weights	IoT network IDS	No	No	N-BaIoT	n/a
StatAvg 2025 [16]	Supervised	FL/mean+var	IoT/IIoT IDS	No	No	TON-IoT, CIC	n/a
Shrivastava 2026 [13]	Semi-sup (M-OCC)	Centralized	IoT sensor anomaly	No	No	IoT sensor	n/a
Belarbi 2023 [11]	Supervised DNN/DBN	FL/weights	IoT IDS	No	No	IoT	n/a
DOF-ID 2023 [17]	Semi-sup (G-net)	Dec. FL	Network IDS	Yes	No	NSL-KDD, BoT	n/a
Attota 2021 [15]	Supervised	FL/weights	IoT IDS	No	Yes	MQTT-IoT	n/a
Wu 2025 [20]	Supervised	None	IoT data streams	Yes	Yes	NSL-KDD	n/a
AF-BKM (ours)	Semi-sup novelty	FL/statistics	Federated IoT edge IDS	No †	No *	NSL-KDD, UNSW-NB15, N-BaIoT	Yes

Each family in Table 2 trades a different strength for a different blind spot. Weight-sharing federated detectors such as FedMSE [14] and the DNN/DBN systems of Belarbi et al. [11] reach strong accuracy and can reuse mature aggregation rules, but the exchanged updates carry full parameter tensors far larger than a fixed $d+3$-value summary, and neither work studies merge-time threshold calibration: threshold selection stays local and manual. Statistics-only designs, including StatAvg [16] and the original BKM [9], suit constrained nodes because a mean vector costs a few hundred bytes, yet their aggregation is fragile: BKM's nearest-anomaly tightening degrades under the very heterogeneity it is meant to absorb (Section 2). Decentralized and online systems [17,20] handle drift well but assume models that do not fit the smallest devices. Ensemble approaches, including the BKM ensemble extension [10], suppress false positives after the fact, at the cost of extra detectors per node, and they inherit whatever miscalibration the base detector accumulates. Mahalanobis one-class methods [13] share our detector's simplicity but are centralized. The literature therefore offers robust aggregation for heavy models or light statistics with brittle aggregation. E1 and E2 target the missing combination: robust, calibrated aggregation for a statistics-only detector.

4. The AF-BKM Method

4.1. Assumptions

AF-BKM is *label-free with respect to attack labels*: it never uses attack labels for thresholding or for worker-quality weighting. It does assume, as in any semi-supervised novelty detector, a small, *trusted, benign* anchor for benign-only calibration. The covariance is estimated only at the coordinator, from the trusted benign anchor, and is never re-estimated from worker data; its inverse is broadcast with the global state so that workers can score locally. Each worker uploads a $d+3$-value summary (its accepted-sample mean plus three scalars: an accepted-sample count, a dispersion value and a local threshold candidate), never raw data and never the covariance. That summary is computed over the samples the current global detector accepts ($\mathcal{A}_k = \{x : D_M(x) \leq \tau\}$), not over ground-truth benign samples: no worker reads a label at any point during detection. In operation, benign labels are needed by the coordinator only to assemble the commissioning anchor; in the experiments, labels are additionally used outside the detector to construct label-skew partitions and compute evaluation metrics. Figure 3 summarizes the federated context inherited from the prior BKM line and marks where AF-BKM modifies the round.

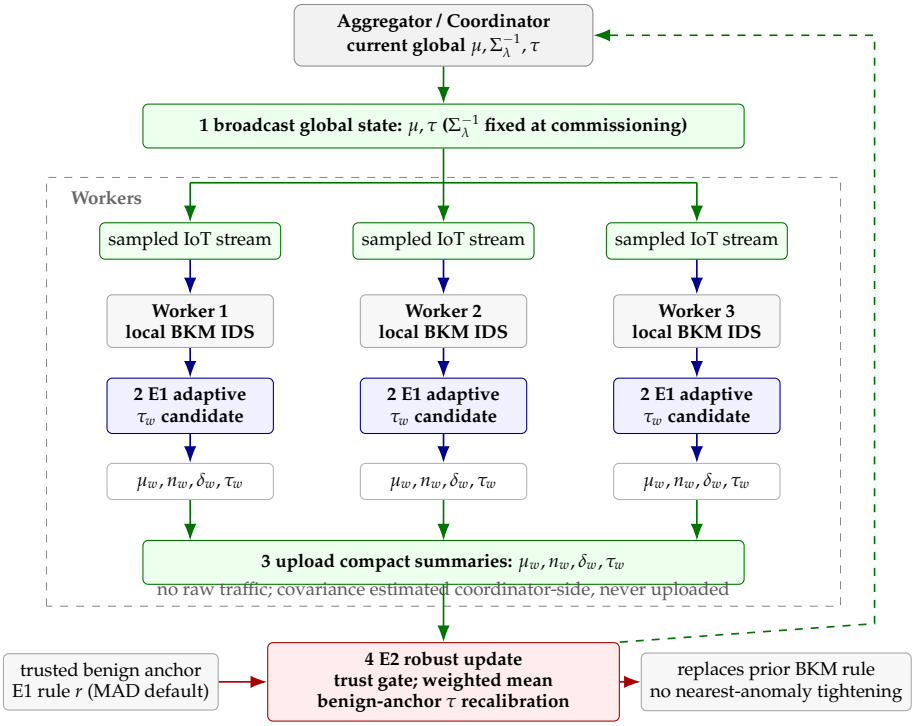


Figure 3. Federated AF-BKM round. The prior coordinator–worker setting is preserved, but the update loop is repaired: workers receive the global state, run local BKM with E1 adaptive threshold candidates, and upload only compact statistics. The coordinator then applies E2 (trust-gated mean fusion and benign-anchor recalibration) instead of the prior nearest-anomaly threshold tightening. Arrow colours denote the type of step: green for communication between the coordinator and the workers, blue for computation local to a worker, and red for the coordinator-side operations introduced by E2.

4.2. Threat Model

The anchor deserves a precise statement because it is the natural target for an adversary. The anchor set (X_a) is collected once, during device commissioning, and is never refreshed from live traffic: after initialization, the coordinator only reads it. Under this threat model, which assumes the coordinator stores the anchor securely and is not, itself, compromised, operational traffic cannot modify the anchor after commissioning. Worker-reported values

do not directly set the threshold: τ is recalibrated on the anchor *points* (X_a; Equation (8)), so a fabricated threshold candidate influences only whether its sender passes the trust gate. The reported mean is the weaker surface, and it is weaker than the threshold rule alone suggests. Equation (8) scores X_a under the *blended* centroid of Equation (7), so a fabricated mean displaces μ and thereby moves τ indirectly, even though no worker value is ever read as a threshold. Beyond that indirect path, β bounds the contribution of the *aggregate* worker block relative to the anchor, but it bounds neither an individual worker's normalized weight, which is computed from that worker's own reported count and dispersion (Equation (6)), nor the magnitude of the reported mean vector; manipulation of the reported count and dispersion is not evaluated in this study. The stress tests of Section 6 probe this surface directly. What an adversary can also attack is the commissioning window: if attack traffic is already flowing while X_a is being collected, the anchor is contaminated before the protocol starts. Section 6 measures both surfaces: targeted faulty updates from compromised workers and contamination of the anchor at collection time.

4.3. E1: Adaptive Auto-Thresholding

Instead of a hand-tuned percentile, the threshold is read from the benign-distance distribution ($\mathcal{D} = \{D_M(x_i)\}$) using a robust MAD rule, with a target-FPR calibration as an alternative:

$$\tau_{\text{MAD}} = \text{median}(\mathcal{D}) + \kappa \cdot 1.4826 \cdot \text{MAD}(\mathcal{D}), \qquad \tau_{\text{FPR}} = \text{quantile}(\mathcal{D}, 1 - \phi), \tag{5}$$

with a default of $\kappa = 3$ and benign target FPR of ϕ. Both are free of a searched percentile. A third search-free rule appears in the standalone threshold study of Section 6 as a comparator: Otsu's between-class variance criterion [24], originally a gray-level image histogram method, applied here to the one-dimensional histogram of benign distances (256 bins) by splitting it where the between-class variance is largest. Otsu is evaluated only as a standalone alternative; the federated pipeline uses the MAD rule by default and the target-FPR rule where an explicit operating point is wanted. E1 is used in two ways: as the *standalone* threshold of a single detector, and in the federated setting, each worker reports its scalar candidate (τ_w) computed by Equation (5), which the coordinator uses only to identify and discard outlier workers (the trust gate below). The final global threshold is calibrated on the trusted anchor (E2), not by worker consensus.

4.4. E2: Robust, Benignly Anchored Aggregation

At each merge, the coordinator (Algorithm 1): (i) discards workers whose threshold candidate is a robust outlier (a trust gate at κ_t scaled MADs from the median of candidates), (ii) forms a count/quality-weighted worker mean and blends it with the trusted anchor mean (μ_a) under a damping factor (β) and (iii) recalibrates the threshold on the trusted anchor (X_a) via Equation (5):

$$w_k \propto n_k/(1 + \delta_k), \qquad \bar{\mu}_{\text{w}} = \sum_{k \in K} w_k \, \mu_k, \tag{6}$$

$$\mu \leftarrow (1 - \beta)\, \mu_a + \beta\, \bar{\mu}_{\text{w}}, \tag{7}$$

$$\tau \leftarrow r\big(\{D_M(x) : x \in X_a\}\big), \tag{8}$$

where K is the set of trusted workers; n_k, δ_k are the size and distance dispersion of worker k's *accepted* set ($\mathcal{A}_k = \{x \in \text{window}_k : D_M(x) \leq \tau\}$), that is, the samples the current global detector accepts as benign, not ground-truth benign samples; β is a blend factor (default 0.5); and r is the E1 rule. The covariance is never uploaded by workers and never re-estimated from their data; the coordinator fixes Σ_λ^{-1} once from the trusted benign anchor

and broadcasts it for local scoring. Because X_a is clean and fixed, τ in Equation (8) tracks a stable, benign FPR across merges, which is what stops the decay, in contrast to the repeated tightening produced by Equation (4) in the evaluated protocol. A side benefit is reduced sensitivity to threshold-inflation poisoning: a malicious worker cannot raise the global threshold directly because τ is never taken from worker-reported values (Section 6). The reduction is partial rather than complete. Equation (8) evaluates the anchor points under the blended μ of Equation (7), so a worker that misreports its mean still shifts τ indirectly. The blend factor (β) scales the aggregate worker block but does not impose an absolute bound because the reported mean magnitude is unbounded.

Algorithm 1: AF-BKM robust benignly anchored merge (E2; one merge round)

Input: trusted anchor X_a, μ_a; inverse covariance Σ_λ^{-1}; worker updates $\{(\mu_k, n_k, \delta_k, \tau_k)\}$; blend β; trust κ_t; E1 rule r
Output: updated global model $(\mu, \Sigma_\lambda^{-1}, \tau)$
$V \leftarrow \{k : \mu_k$ exists and $n_k > 0\}$;
if $V = \emptyset$ **then return** previous $(\mu, \Sigma_\lambda^{-1}, \tau)$ unchanged;
$C \leftarrow \{k \in V : \tau_k$ is finite$\}$;
if $|C| < 3$ **then**
  $K \leftarrow V$; `// gate abstains`
else
  $\tilde{m} \leftarrow \text{median}(\{\tau_k : k \in C\})$; $s \leftarrow 1.4826\,\text{MAD}(\{\tau_k : k \in C\})$;
  $K \leftarrow \{k \in C : |\tau_k - \tilde{m}| \le \kappa_t s\}$; `// trust gate: drop outlier workers`
  if $K = \emptyset$ **then** $K \leftarrow V$ `// gate abstains;`
if $K = \emptyset$ **then return** previous $(\mu, \Sigma_\lambda^{-1}, \tau)$ unchanged;
$w_k \leftarrow n_k/(1+\delta_k)$ for $k \in K$; normalize w;
$\bar{\mu}_w \leftarrow \sum_{k \in K} w_k \mu_k$; $\mu \leftarrow (1-\beta)\mu_a + \beta \bar{\mu}_w$; `// Equations (6) and (7)`
$\tau \leftarrow r(\{D_M(x) : x \in X_a\})$; `// benign-anchored recalibration, Equation (8)`
return $(\mu, \Sigma_\lambda^{-1}, \tau)$

5. Experimental Setup

Datasets and baseline-only scaling. We use three public datasets. NSL-KDD [25] and UNSW-NB15 [26] are the standard network-IDS benchmarks; we pool their official partitions to form the federated stream and keep the continuous numeric features after dropping near-constant columns ($d = 37$ and $d = 39$; 148,517 and 257,673 records; attack ratios 0.48 and 0.64). The third dataset, N-BaIoT [27], is newer than both benchmarks and IoT-specific: controlled traffic captures from nine commercial IoT devices (doorbells, thermostats, baby monitors, cameras, a webcam) recorded before and during Mirai and BASHLITE botnet infections, summarized as 115 numeric stream statistics per packet. We use the prepared 50-client non-IID partition distributed with the FedMSE code release (pinned to commit `d126885`), pooling each client's two benign partitions and its attack partition into one corpus ($d = 115$; 136,565 records; 71,716 benign and 64,849 attack, ratio 0.47); the experiments then repartition this pool with the seeded Dirichlet protocol described below, like the other two datasets. To avoid scaler leakage, the MinMax scaler is fit *only on the coordinator's benign baseline* and then applied to the (future) stream. One preprocessing step is different: the near-constant column filter (retaining columns with standard deviation above 10^{-8}) is computed once over the pooled feature matrix before the seeded splits. This filter reads features but not labels; recomputing it from all benign rows produced the same retained-column mask on all three corpora (NSL-KDD drops only

num_outbound_cmds; UNSW-NB15 and N-BaIoT drop nothing). We therefore report it as label-blind full-corpus preprocessing rather than as a strictly inductive baseline-only step. We pool the official partitions rather than evaluating on the canonical train/test splits; an official-split sensitivity study is left to future work.

Original-baseline fidelity. The comparison baseline ("Original") is a protocol-level reimplementation of the published coordinator equations, Equations (2)–(4), evaluated inside the same windowed simulation as AF-BKM: worker updates are formed from the current stream window, whereas the original released implementation accumulates per-worker distance histories across the run. Both aggregation modes share this current-window convention, so the comparison isolates the merge rules themselves.

Federated protocol. A coordinator initializes a benign baseline of 2000 packets and shares statistics with W workers. In a real deployment the workers rarely observe the same traffic: each IoT node sees a device-specific mix of benign and attack flows, so their local data are *non-independent and identically distributed* (non-IID), and the merge rule must stay accurate when the workers' distributions disagree. We emulate this by partitioning the remaining records across workers with a Dirichlet(α) label-skew, where a smaller α yields a more skewed, more non-IID split of the benign/attack labels (a small uniform floor keeps every worker viable); the resulting stream is processed in sliding windows, with merges at fixed epochs. Online classification of streaming sensor incidents has been studied in related monitoring contexts [28]. Defaults are listed in Table 3. Confidence intervals are percentile bootstrap intervals over the paired per-seed differences (10,000 resamples, fixed seed). The central ablation and the significance tests use 10 seeds; the secondary analyses (sweeps, target-FPR, contamination, faulty-worker, evasion) use 5 seeds and the detector and resource comparisons 3, as stated with each experiment; all results report mean ± std.

Table 3. Default hyperparameters (one compact reference). Sweeps vary α and W.

Parameter	Value	Parameter	Value
Benign baseline $\lvert X_a \rvert$	2000	Window size	1000
Workers W	3	Merges	3 (epochs 6, 12, 18)
LLLL Non-IID α	0.5	Blend β	0.5
Trust gate κ_t	3	MAD factor κ	3
Ridge λ	10^{-3}	Seeds	10

Ablation factors. Aggregation (*original* vs. *robust* = E2) × threshold (*fixed percentile* $p = 90$ vs. *adaptive* MAD = E1): *Baseline*, *E1*, *E2*, *AF-BKM (E1 + E2)*.

Baselines and metrics. We compare the Mahalanobis core against One-Class SVM, Isolation Forest, Local Outlier Factor, Kernel Density Estimation and a k-NN detector [29]. For fairness, all detectors are trained on the *same* 2000 benign packets with fixed a priori settings chosen without attack labels (One-Class SVM: RBF kernel, $\nu = 0.1$; Isolation Forest: 150 trees; LOF: 50 neighbors; KDE: bandwidth 0.5; k-NN: $k = 5$; benign-quantile cutoffs at 5% where the detector needs one); the full configuration is in the released repository. We report precision, recall, F1, the recall-weighted F2, FPR, and the change in precision ΔP from the first to the last three epochs (negative = decay).

6. Results and Analysis

6.1. Ablation: E2 Removes the Precision Decay

Figure 4 and Table 4 report the central result. Under the original merge, precision falls steadily after each merge, and the FPR climbs on all three datasets ($\Delta P = -0.134$ on NSL-KDD, -0.121 on UNSW-NB15, and -0.170 on N-BaIoT, the strongest decay of

the three), reproducing the published failure mode. Changing only the threshold rule (E1) does not fix it ($\Delta P = -0.119$ on NSL-KDD), confirming that the *aggregation*, not the threshold rule, drives the decay. Robust aggregation (E2 and AF-BKM) flattens the precision curve ($\Delta P \approx -0.002$ on NSL-KDD; $-0.006/-0.014$ on UNSW-NB15; -0.009 on N-BaIoT) and reduces the FPR by 30–64% (NSL-KDD $0.208 \rightarrow 0.108$–0.145; UNSW-NB15 $0.219 \rightarrow 0.079$–0.104; N-BaIoT $0.210 \rightarrow 0.102$–0.106). Recall behaves differently per dataset: NSL-KDD preserves it ($0.94 \rightarrow 0.92$ for AF-BKM); N-BaIoT saturates at 1.000 throughout (the Mirai/BASHLITE flows sit far from benign traffic in the 115-dimensional feature space, so the entire contest is about false positives, which is exactly where the original merge loses ground); and on the harder UNSW-NB15, the MAD operating point trades recall for precision, which we make explicit in Section 6.4. Table 5 confirms the central claims are statistically significant across 10 seeds on all three datasets: every comparison reaches the minimum two-sided Wilcoxon $p = 0.002$ (AF-BKM better on all seeds) with large effect sizes (d_z from $+2.2$ to $+6.0$) and bootstrap 95% CIs that exclude zero; on N-BaIoT, the final-precision improvement is the largest of the three ($+0.157$).

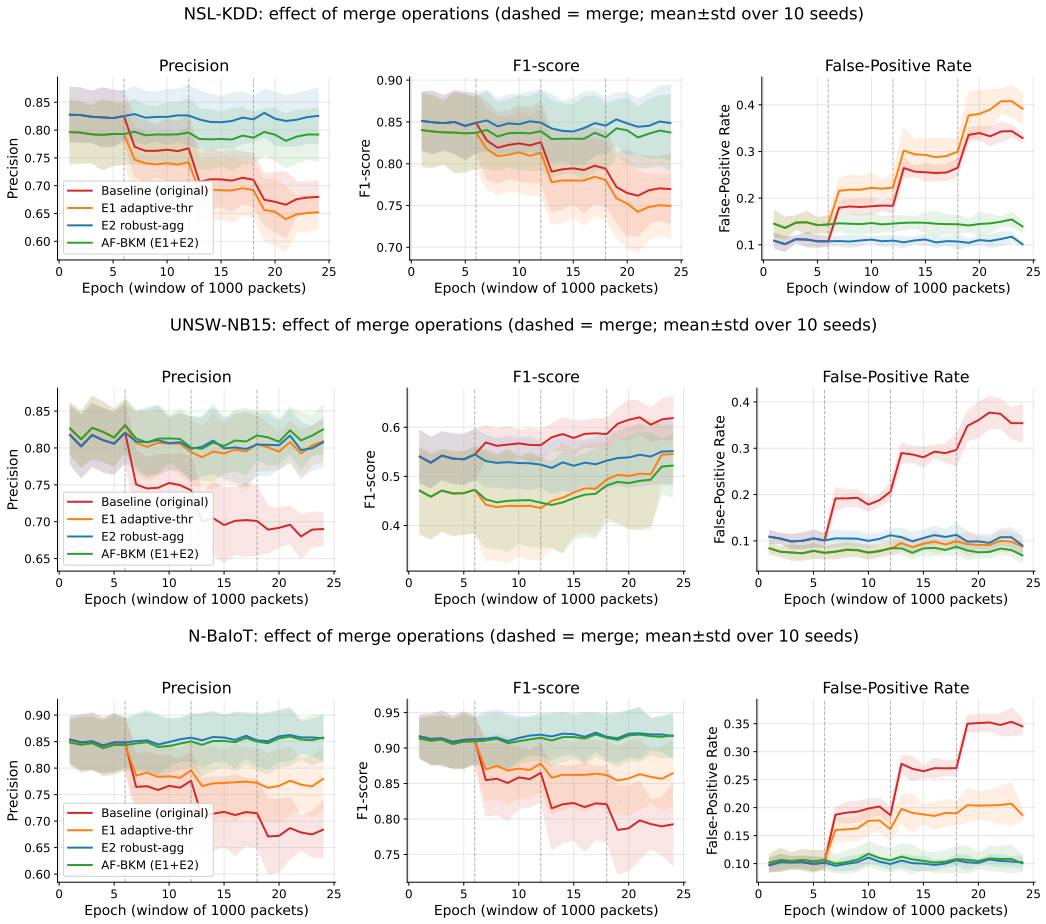


Figure 4. Effect of merge operations (dashed lines) on NSL-KDD (**top**), UNSW-NB15 (**middle**) and N-BaIoT (**bottom**)—mean ± std over 10 seeds. Baseline/E1 (original merge) lose precision and gain FPR after each merge; E2/AF-BKM (robust aggregation) keep precision flat at a low, controlled FPR.

Table 4. Ablation over the two mechanisms on NSL-KDD, UNSW-NB15 and N-BaIoT (3 workers, 3 merges, mean ± std over 10 seeds). ΔP is the change in precision from the first to the last three epochs; negative indicates merge-induced decay.

Dataset	Config	Precision (Final)	ΔP (Early→Late)	Recall	F1	F2	FPR
NSL-KDD	Baseline (original)	0.692 ± 0.067	−0.134	0.940 ± 0.011	0.813 ± 0.044	0.874 ± 0.028	0.208 ± 0.030
NSL-KDD	E1 adaptive-thr	0.675 ± 0.072	−0.119	0.954 ± 0.012	0.801 ± 0.049	0.872 ± 0.031	0.247 ± 0.046
NSL-KDD	E2 robust-agg	0.824 ± 0.049	−0.002	0.898 ± 0.006	0.847 ± 0.036	0.873 ± 0.021	0.108 ± 0.006
NSL-KDD	AF-BKM (E1+E2)	0.792 ± 0.053	−0.002	0.923 ± 0.008	0.836 ± 0.043	0.879 ± 0.026	0.145 ± 0.016
UNSW-NB15	Baseline (original)	0.692 ± 0.041	−0.121	0.494 ± 0.070	0.570 ± 0.037	0.516 ± 0.055	0.219 ± 0.027
UNSW-NB15	E1 adaptive-thr	0.793 ± 0.040	−0.029	0.343 ± 0.100	0.467 ± 0.100	0.383 ± 0.102	0.085 ± 0.012
UNSW-NB15	E2 robust-agg	0.806 ± 0.040	−0.006	0.410 ± 0.067	0.532 ± 0.049	0.450 ± 0.061	0.104 ± 0.011
UNSW-NB15	AF-BKM (E1+E2)	0.808 ± 0.037	−0.014	0.338 ± 0.091	0.465 ± 0.091	0.378 ± 0.092	0.079 ± 0.008
N-BaIoT	Baseline (original)	0.681 ± 0.057	−0.170	1.000 ± 0.000	0.854 ± 0.044	0.930 ± 0.026	0.210 ± 0.041
N-BaIoT	E1 adaptive-thr	0.757 ± 0.056	−0.089	1.000 ± 0.000	0.879 ± 0.039	0.944 ± 0.021	0.159 ± 0.028
N-BaIoT	E2 robust-agg	0.842 ± 0.048	−0.009	1.000 ± 0.000	0.915 ± 0.036	0.962 ± 0.019	0.102 ± 0.007
N-BaIoT	AF-BKM (E1+E2)	0.838 ± 0.049	−0.009	1.000 ± 0.000	0.913 ± 0.034	0.961 ± 0.017	0.106 ± 0.017

Table 5. Paired comparison of AF-BKM vs. the original baseline across 10 seeds. Improvement is the mean paired difference (FPR oriented as a reduction) with a bootstrap 95% CI; significance by Wilcoxon signed-rank; effect size as Cohen's d_z.

Dataset	Metric	Baseline	AF-BKM	Improvement [95% CI]	Wilcoxon p	Cohen dz
NSL-KDD	Final precision	0.692 ± 0.067	0.792 ± 0.053	+0.100 [+0.075, +0.124]	0.0020	+2.37
NSL-KDD	Mean FPR	0.203 ± 0.029	0.144 ± 0.015	+0.059 [+0.043, +0.073]	0.0020	+2.23
NSL-KDD	Precision decay ΔP	−0.134 ± 0.041	−0.002 ± 0.014	+0.132 [+0.103, +0.158]	0.0020	+2.82
UNSW-NB15	Final precision	0.692 ± 0.041	0.808 ± 0.037	+0.116 [+0.094, +0.137]	0.0020	+3.19
UNSW-NB15	Mean FPR	0.215 ± 0.025	0.078 ± 0.008	+0.137 [+0.123, +0.149]	0.0020	+5.99
UNSW-NB15	Precision decay ΔP	−0.121 ± 0.016	−0.014 ± 0.015	+0.107 [+0.094, +0.119]	0.0020	+5.16
N-BaIoT	Final precision	0.681 ± 0.057	0.838 ± 0.049	+0.157 [+0.135, +0.178]	0.0020	+4.30
N-BaIoT	Mean FPR	0.199 ± 0.039	0.105 ± 0.016	+0.094 [+0.070, +0.115]	0.0020	+2.44
N-BaIoT	Precision decay ΔP	−0.170 ± 0.039	−0.009 ± 0.007	+0.162 [+0.135, +0.184]	0.0020	+3.81

6.2. E1: Removing the Manual Threshold

Figure 5 and Table 6 motivate the adaptive threshold. The F1 of a fixed-percentile detector varies strongly with the chosen percentile (over $[0.781, 0.892]$ on NSL-KDD, $[0.307, 0.726]$ on UNSW-NB15 and $[0.858, 0.987]$ on N-BaIoT), so the "right" percentile is dataset-dependent and unknowable a priori. The three adaptive rules compared here (MAD, Otsu and the calibrated quantile) are search-free rather than parameter-free: MAD uses a fixed $\kappa = 3$, Otsu uses a fixed bin count, and the calibrated rule uses an operator-chosen target FPR; none uses attack labels. On NSL-KDD and N-BaIoT, they land close to the best manual percentile (0.887 and 0.972 against best-manual 0.892 and 0.987). On UNSW-NB15, they provide a stable, controlled operating point but do not reach the best manual choice (0.557 for Otsu, the best adaptive rule there, against 0.726 for the best percentile): with hindsight, an oracle percentile buys recall that a benign-only rule cannot see, which is the same narrow-margin behavior that reappears in the operating-point and evasion analyses.

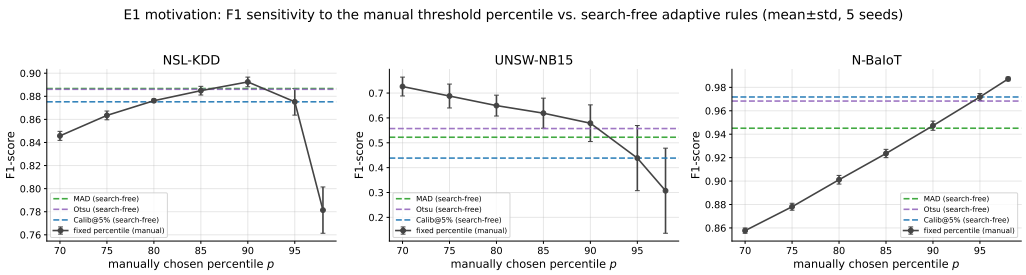


Figure 5. Standalone-detector F1 versus the manually chosen percentile (solid) against search-free adaptive rules (dashed)—mean ± std over 5 seeds.

Table 6. Standalone-detector F1 under manual percentile selection vs. search-free adaptive thresholds. The manual percentile must be searched per dataset; the adaptive rules land near the best manual percentile on NSL-KDD and N-BaIoT, whereas on UNSW-NB15, they provide a controlled operating point but do not reach the best manual choice (0.557 vs. 0.726).

Dataset	Best Fixed-p F1	F1 Range over p	MAD F1	Otsu F1	Calib@5% F1
NSL-KDD	0.892 (p = 90)	0.781–0.892	0.887	0.886	0.875
UNSW-NB15	0.726 (p = 70)	0.307–0.726	0.522	0.557	0.438
N-BaIoT	0.987 (p = 98)	0.858–0.987	0.945	0.968	0.972

6.3. Robustness Across Heterogeneity and Scale

Figure 6 sweeps the non-IID severity (α) and the number of workers. Across every setting and all three datasets, AF-BKM keeps a higher and more stable final precision than the Baseline (e.g., on UNSW-NB15 at $\alpha = 5$, 0.885 vs. 0.756; with 10 workers, 0.793 vs. 0.675; on N-BaIoT at $\alpha = 5$, 0.891 vs. 0.742, and with 10 workers, 0.813 vs. 0.647), with consistently lower FPR; the benefit holds under strong non-IID ($\alpha = 0.1$) and as the federation grows.

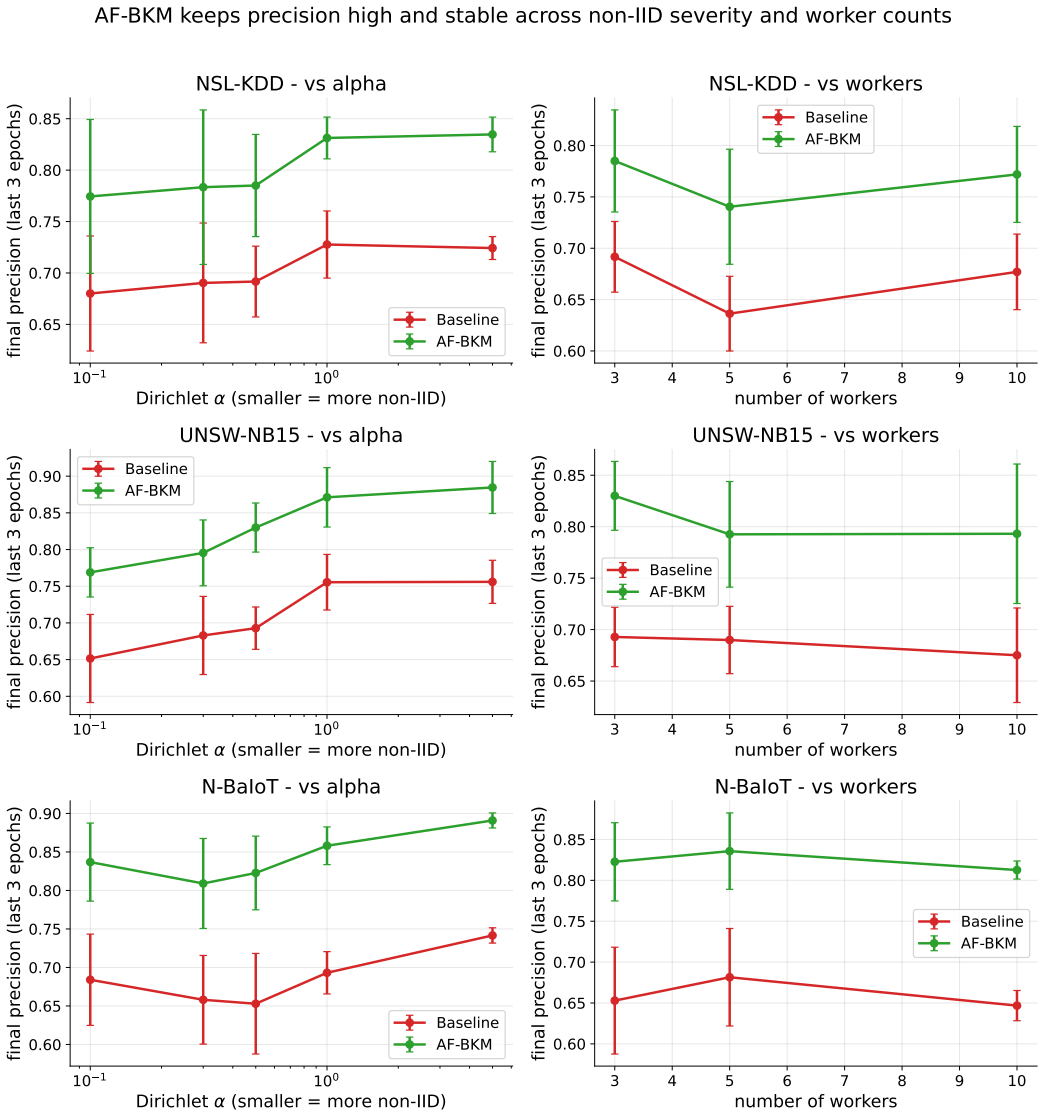


Figure 6. Final precision versus Dirichlet α (**left**) and number of workers (**right**) for NSL-KDD (**top**), UNSW-NB15 (**middle**) and N-BaIoT (**bottom**). AF-BKM remains above the original merge in every plotted setting.

6.4. UNSW-NB15: An Explicit Precision–Recall Operating Point

On UNSW-NB15, attacks are harder to separate from benign traffic, and at the low-FPR operating point the MAD rule selects, AF-BKM favors precision and FPR over recall (Table 4). Because the threshold is benignly anchored to a target FPR (Equation (8) with $r = \tau_{\text{FPR}}$), the operator can choose the trade-off explicitly. Table 7 and Figure 7 sweep the benign target FPR ($\phi \in \{0.05, 0.10, 0.15, 0.20\}$): the achieved FPR tracks ϕ, and on UNSW-NB15, recall rises from 0.28 to 0.53 and F1 from 0.40 to 0.60 as ϕ grows from 0.05 to 0.20, while precision falls gently from 0.83 to 0.74. A mid-range $\phi \approx 0.15$–0.20 is a reasonable operating point on UNSW-NB15; on NSL-KDD, the knob spans a recall range of 0.81–0.95, and on N-BaIoT, recall stays at 1.000 for every ϕ, so the knob simply selects the tolerated false-positive level. We therefore frame AF-BKM as a precision/FPR-stability repair with an explicit, controllable operating point, not as a method that maximizes recall on every dataset.

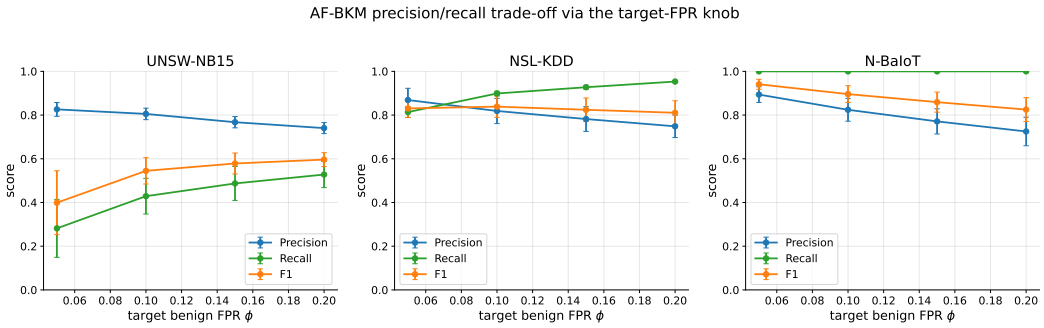


Figure 7. AF-BKM precision/recall/F1 versus the benign target-FPR knob (ϕ) (federated, mean±std over 5 seeds).

Table 7. AF-BKM operating points on the benign target-FPR knob (federated, mean ± std over 5 seeds). The achieved FPR tracks the target, exposing an explicit precision/recall trade-off, most pronounced on UNSW-NB15.

Dataset	Target FPR	Precision	Recall	F1	F2	Actual FPR
UNSW-NB15	0.05	0.826 ± 0.032	0.282 ± 0.133	0.399 ± 0.146	0.318 ± 0.139	0.050 ± 0.010
UNSW-NB15	0.1	0.805 ± 0.026	0.429 ± 0.082	0.545 ± 0.060	0.467 ± 0.075	0.100 ± 0.015
UNSW-NB15	0.15	0.767 ± 0.026	0.487 ± 0.078	0.579 ± 0.048	0.516 ± 0.066	0.156 ± 0.015
UNSW-NB15	0.2	0.740 ± 0.025	0.528 ± 0.060	0.596 ± 0.032	0.548 ± 0.047	0.208 ± 0.010
NSL-KDD	0.05	0.869 ± 0.054	0.814 ± 0.025	0.830 ± 0.041	0.817 ± 0.031	0.057 ± 0.004
NSL-KDD	0.1	0.819 ± 0.058	0.899 ± 0.011	0.839 ± 0.049	0.866 ± 0.032	0.110 ± 0.008
NSL-KDD	0.15	0.782 ± 0.057	0.927 ± 0.009	0.824 ± 0.054	0.871 ± 0.037	0.160 ± 0.010
NSL-KDD	0.2	0.749 ± 0.051	0.953 ± 0.004	0.811 ± 0.055	0.874 ± 0.041	0.216 ± 0.006
N-BaIoT	0.05	0.894 ± 0.036	1.000 ± 0.000	0.941 ± 0.023	0.974 ± 0.011	0.054 ± 0.006
N-BaIoT	0.1	0.824 ± 0.052	1.000 ± 0.000	0.896 ± 0.038	0.952 ± 0.021	0.102 ± 0.008
N-BaIoT	0.15	0.771 ± 0.057	1.000 ± 0.000	0.859 ± 0.046	0.933 ± 0.027	0.151 ± 0.007
N-BaIoT	0.2	0.725 ± 0.065	1.000 ± 0.000	0.825 ± 0.055	0.914 ± 0.035	0.202 ± 0.010

6.5. The Mahalanobis Core Is Efficient but Not Sufficient Alone

Table 8 and Figure 8 compare the standalone Mahalanobis core against classical novelty detectors on the same 2000 benign packets. On NSL-KDD, it attains the best recall (0.92) and an F1 of 0.90, on par with KDE and k-NN, while being the cheapest to fit and run (4 ms fit; 35 ms to score a 20,000-packet batch in one vectorized call, versus approximately 150 ms and 1.5 s for KDE). On N-BaIoT, it reaches an F1 of 0.956 with full recall; KDE edges it on F1 (0.978), at roughly ten times the fit cost and seven times the scoring cost. On UNSW-NB15, however, the same core alerts sparingly: precision remains high (0.924), and

FPR is moderate (0.082), but recall falls to 0.425 and F1 to 0.580, below k-NN and LOF. The result should therefore be read as motivation rather than as a standalone victory: BKM is an efficient statistics-only base layer, but on harder traffic, it needs the adaptive thresholding and robust federated recalibration introduced by AF-BKM.

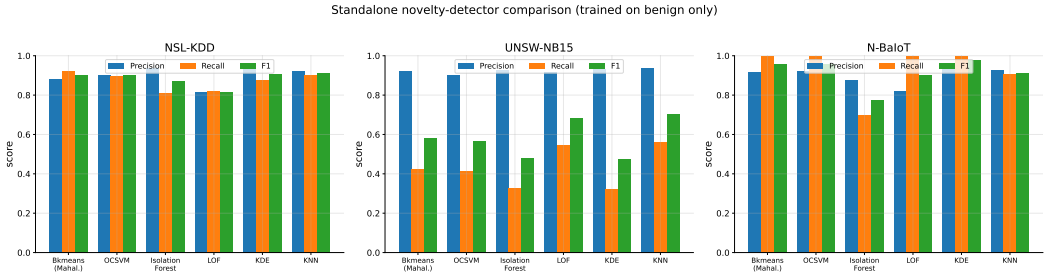


Figure 8. Standalone novelty-detector comparison (trained on benign traffic only), mean over three seeds.

Table 8. Standalone semi-supervised novelty detectors trained on 2000 benign samples and evaluated on a 20k-packet stream (mean over 3 seeds). The Mahalanobis core is efficient and competitive on NSL-KDD; its low recall on UNSW-NB15 exposes the limit of standalone BKM and motivates AF-BKM's adaptive thresholding and robust aggregation.

Dataset	Detector	Precision	Recall	F1	F2	FPR	Fit (ms)	Predict (ms)
N-BaIoT	Bkmeans–Mahalanobis	0.915	1.000	0.956	0.982	0.102	40	372
N-BaIoT	IsolationForest	0.875	0.698	0.777	0.727	0.109	69	52
N-BaIoT	KDE	0.958	1.000	0.978	0.991	0.049	367	2764
N-BaIoT	KNN	0.930	0.907	0.913	0.908	0.074	7	19
N-BaIoT	LOF	0.819	1.000	0.900	0.958	0.242	12	28
N-BaIoT	OCSVM	0.921	1.000	0.959	0.983	0.094	16	228
NSL-KDD	Bkmeans–Mahalanobis	0.881	0.924	0.902	0.915	0.140	4	35
NSL-KDD	IsolationForest	0.938	0.812	0.870	0.834	0.060	69	57
NSL-KDD	KDE	0.945	0.876	0.909	0.889	0.058	147	1482
NSL-KDD	KNN	0.924	0.902	0.913	0.906	0.084	5	11
NSL-KDD	LOF	0.815	0.819	0.817	0.818	0.209	19	24
NSL-KDD	OCSVM	0.904	0.896	0.900	0.898	0.107	9	121
UNSW-NB15	Bkmeans–Mahalanobis	0.924	0.425	0.580	0.475	0.082	4	38
UNSW-NB15	IsolationForest	0.927	0.325	0.481	0.374	0.062	67	59
UNSW-NB15	KDE	0.930	0.322	0.474	0.369	0.053	159	1584
UNSW-NB15	KNN	0.938	0.562	0.702	0.611	0.088	5	11
UNSW-NB15	LOF	0.917	0.545	0.683	0.593	0.118	11	24
UNSW-NB15	OCSVM	0.904	0.415	0.567	0.464	0.103	9	123

6.6. Communication Cost and Selected Faulty-Worker Update Stress Tests

AF-BKM's per-worker uplink is a d+3-value fp32 payload per merge round: one benign mean plus a benign count, a dispersion value and a threshold candidate. This amounts to 160, 168 and 472 bytes on the three datasets; extending the same payload with the covariance upper triangle ($d(d+1)/2$ additional values) would multiply it by 18–58, so the statistics-only design saves 94.6–98.3% of the comparison uplink (Table 9). The downlink is smaller still. Σ_λ^{-1} is estimated once on the commissioning anchor and returned unchanged by every merge (Algorithm 1), so broadcasting it is a one-time initialization cost already counted in the 5.5–52 KB deployed model of Section 6.9; the recurring per-round broadcast is only (μ, τ), that is, d+1 fp32 values (152–464 B). The bidirectional per-round cost is therefore about twice the reported uplink, not orders of magnitude above it. For robustness, we stress the aggregation with workers that deviate from the protocol in targeted, scripted ways, in the spirit of the faulty-update literature of federated learning [30,31]. Five manipulations are evaluated: *evasion* (the worker declares its whole

window benign and inflates its candidate), *mean-shift* (the worker reports the attack-region centroid as its benign mean behind an honest-looking candidate), *threshold inflation* and *threshold deflation* (honest mean, fabricated candidate in either direction), each with one of three workers compromised, plus evasion with two of five. Table 10 and Figure 9 report honest-worker precision and recall on NSL-KDD. Under every tested manipulation, AF-BKM retains higher honest-worker *precision* than the original merge (evasion 1/3: 0.73 vs. 0.61); its recall is lower than the original merge in each of these scenarios, consistent with the low-FPR operating point it selects. A fabricated threshold candidate does not set τ because the threshold is recalibrated on the anchor; the candidate only affects whether its sender passes the MAD trust gate. The reported mean is the harder surface, and the protection there is weaker than the blend factor alone suggests. It is also the one path by which a worker still reaches the threshold: Equation (8) scores the anchor points under the blended centroid, so displacing μ displaces τ with it, which is visible in the mean-shift row. The blend factor (β) bounds the contribution of the *aggregate* worker block relative to the trusted anchor; it does not bound an individual worker's normalized weight because the weights in Equation (6) are computed from the worker's own reported count (n_k) and dispersion (δ_k), nor does it bound the magnitude of the reported mean vector. A worker that inflates its count or understates its dispersion therefore claims a larger share of the worker block, and the gate does not inspect either quantity; manipulation of the reported count and dispersion is not evaluated here. The mean-shift results show measurable degradation that is mitigated but not neutralized. These experiments establish behavior under the five scripted manipulations only, not worst-case robustness: an adaptive adversary could optimize its update against the defense [32], and defending the vector update itself (for example, norm clipping combined with a robust vector aggregator) is a method extension we defer to future work.

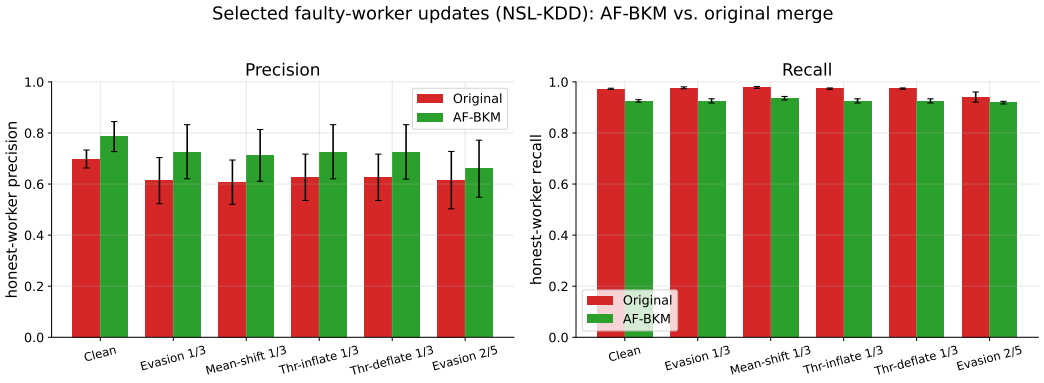


Figure 9. Honest-worker precision and recall under the selected faulty-worker updates (NSL-KDD): AF-BKM vs. the original merge, mean ± std over 5 seeds.

Table 9. Per-worker uplink per merge round (fp32). AF-BKM transmits one benign mean, a benign count, a dispersion value, and a threshold candidate ($d + 3$ values); sharing the covariance adds $d(d + 1)/2$ values.

Dataset	d	Statistics-Only (Floats)	Share Covariance (Floats)	Uplink Reduction	Stats (B/round)	Cov (B/round)
NSL-KDD	37	40	743	94.6%	160	2972
UNSW-NB15	39	42	822	94.9%	168	3288
N-BaIoT	115	118	6788	98.3%	472	27,152

Table 10. Honest-worker precision (P) and recall (R) under selected faulty-worker updates (mean over 5 seeds). Fabricated threshold candidates (inflation and deflation) do not directly set AF-BKM's threshold; mean-shift, which targets the blended centroid, is mitigated but not neutralized.

Scenario	Orig. P	Orig. R	AF-BKM P	AF-BKM R
Clean	0.698	0.973	0.786	0.925
Evasion 1/3	0.613	0.977	0.726	0.926
Mean-shift 1/3	0.607	0.978	0.712	0.936
Thr-inflate 1/3	0.626	0.974	0.726	0.926
Thr-deflate 1/3	0.626	0.974	0.726	0.926
Evasion 2/5	0.615	0.941	0.660	0.919

6.7. Contamination of the Trusted Anchor

The remaining attack surface identified in the threat model is the commissioning window itself: if attack traffic is already flowing while the anchor is collected, X_a is contaminated before the protocol starts. We measure this directly by replacing a fraction ($p \in \{0, 1, 2, 5, 10, 20\}$%) of the 2000-packet baseline with attack rows before fitting; the contamination then propagates through the scaler, the mean and covariance, the threshold, and the coordinator anchor, as a dirty collection window would. Table 11 and Figure 10 report the measured pattern (NSL-KDD and UNSW-NB15, five seeds). The false-positive rate stays controlled at every contamination level (0.08–0.16 on NSL-KDD, 0.07–0.08 on UNSW-NB15). Recall behaves differently per dataset: on NSL-KDD, it stays within a point of the clean run through $p = 2\%$ ($0.925 \rightarrow 0.913$, with precision moving from 0.79 to 0.84), whereas on UNSW-NB15, where recall is threshold-sensitive to begin with, it already declines over the same range (0.366 at $p = 0$ to 0.287 at $p = 2\%$, with substantial seed-to-seed variation). From $p = 5\%$ recall drops sharply on both datasets (NSL-KDD, 0.45 at 5% and 0.25 at 10%; UNSW-NB15, 0.18 and 0.13). The MAD threshold is a robust statistic and holds the false-positive level, but the mean and covariance underneath it are *not* robust estimators, and the planted attack rows inflate the covariance along attack directions, which shrinks attack distances and costs recall. The threshold rule survives contamination; the Gaussian model is the weak point, which argues for MCD or another robust covariance estimator as a hardening step. The tolerated regime therefore rests on an operational assumption rather than a guarantee: commissioning is expected to be a short, supervised phase, and contamination beyond a few percent of that window falls outside what the current estimators absorb.

Table 11. AF-BKM under commissioning-anchor contamination: a fraction (p) of the nominally benign collection window is replaced with attack traffic (mean $\pm$ std over 5 seeds).

Dataset	Contamination	Precision	Recall	F1	FPR
NSL-KDD	0%	0.786 ± 0.059	0.925 ± 0.005	0.826 ± 0.054	0.155 ± 0.014
NSL-KDD	1%	0.804 ± 0.064	0.921 ± 0.002	0.842 ± 0.055	0.138 ± 0.014
NSL-KDD	2%	0.836 ± 0.047	0.913 ± 0.009	0.863 ± 0.032	0.133 ± 0.015
NSL-KDD	5%	0.714 ± 0.012	0.452 ± 0.055	0.535 ± 0.036	0.123 ± 0.011
NSL-KDD	10%	0.696 ± 0.043	0.245 ± 0.012	0.349 ± 0.019	0.081 ± 0.006
NSL-KDD	20%	0.653 ± 0.062	0.185 ± 0.023	0.271 ± 0.022	0.077 ± 0.016
UNSW-NB15	0%	0.813 ± 0.036	0.366 ± 0.096	0.490 ± 0.093	0.075 ± 0.010
UNSW-NB15	1%	0.777 ± 0.043	0.314 ± 0.062	0.434 ± 0.071	0.073 ± 0.011
UNSW-NB15	2%	0.835 ± 0.045	0.287 ± 0.065	0.417 ± 0.074	0.078 ± 0.013
UNSW-NB15	5%	0.723 ± 0.051	0.177 ± 0.023	0.278 ± 0.033	0.082 ± 0.011
UNSW-NB15	10%	0.706 ± 0.059	0.125 ± 0.019	0.207 ± 0.025	0.078 ± 0.007
UNSW-NB15	20%	0.600 ± 0.040	0.090 ± 0.009	0.149 ± 0.016	0.083 ± 0.013

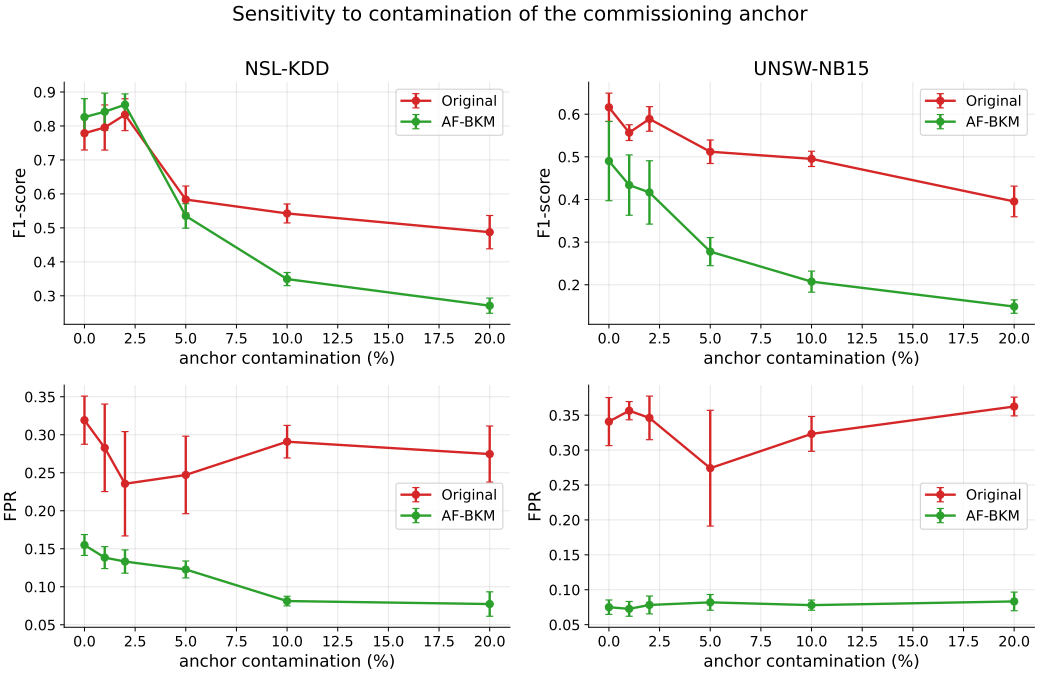


Figure 10. F1 and FPR as a function of the fraction (p) of attack rows planted in the initial trusted anchor (mean $\pm$ std over 5 seeds). The FPR stays controlled for all p values; recall (and hence F1) degrades once contamination exceeds a few percent.

6.8. White-Box Evasion

Beyond faulty workers, an attacker may craft the traffic itself. This analysis is a *detector-level, pre-merge* stress test: it evaluates the detection rule on standalone models fitted on the benign baseline, not the post-merge state of a federated run, so it characterizes the rule that every AF-BKM worker applies rather than one particular federated endpoint. We consider a fully informed attacker who knows the benign centroid (μ) and moves each attack sample toward it ($x' = x + t(\mu - x)$) with budget $t \in [0, 1)$. The Mahalanobis distance is homogeneous around the mean ($D_M(x') = (1 - t)D_M(x)$), so recall under attack follows in closed form, and the three threshold rules differ only through τ. The reported metric is *relative*: for each seed, the budget at which recall falls to half of that seed's own unperturbed recall so that datasets with different clean recall levels are compared on the same footing. Table 12 and Figure 11 report it. On N-BaIoT, the halving budget is 0.98–0.99: botnet flows sit so far from benign traffic that the attacker must make them statistically indistinguishable from it before detection halves. On NSL-KDD, the budget is 0.56–0.71, with the MAD rule the most resistant of the three. On UNSW-NB15, the mean budget is 0.42–0.47 but with very large seed-to-seed variation ($\pm$0.23–0.30): the benign–attack margin there is narrow and unstable, so the cost of mimicry depends heavily on which traffic mix a seed draws. The overall pattern is that the evasion budget tracks the benign–attack margin of each dataset, and where that margin is narrow, complementary defenses, for instance, the ensemble layer of the BKM line or payload-level features, become relevant. A gray-box attacker without μ would first have to estimate it through probing, which the FPR-calibrated threshold makes noisy.

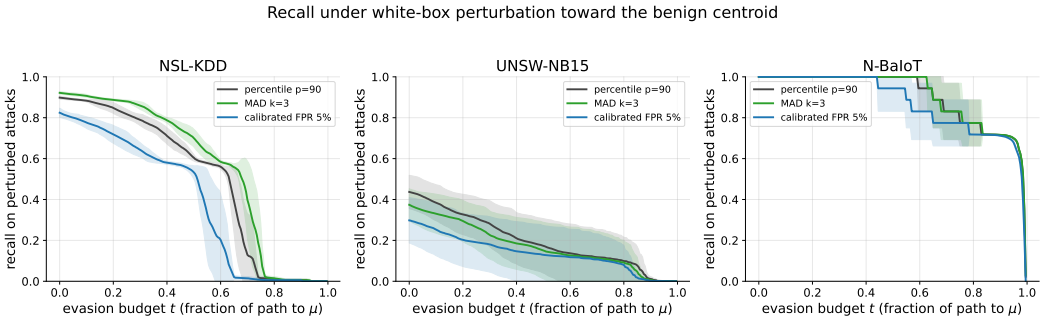


Figure 11. Recall as a fully informed attacker moves attack samples toward the benign centroid with budget t (mean $\pm$ std over 5 seeds).

Table 12. White-box feature-space evasion toward the benign centroid. Budget t is the fraction of the path from an attack sample to μ; the table reports the budget required to halve unperturbed recall (mean $\pm$ std over 5 seeds). Larger is better.

Dataset	Threshold Rule	Initial Recall	Budget to Halve Recall
NSL-KDD	percentile $p = 90$	0.898	0.66 ± 0.03
NSL-KDD	MAD k = 3	0.922	0.71 ± 0.03
NSL-KDD	calibrated FPR 5%	0.824	0.56 ± 0.05
UNSW-NB15	percentile $p = 90$	0.437	0.47 ± 0.23
UNSW-NB15	MAD k = 3	0.374	0.47 ± 0.26
UNSW-NB15	calibrated FPR 5%	0.298	0.42 ± 0.30
N-BaIoT	percentile $p = 90$	1.000	0.99 ± 0.00
N-BaIoT	MAD k = 3	1.000	0.99 ± 0.00
N-BaIoT	calibrated FPR 5%	1.000	0.98 ± 0.00

6.9. Computational Resource Profile

Table 13 profiles the worker-side detector on the three datasets, and Table 14 compares it with the classical detectors. All timings are wall-clock desktop benchmarks on the stated CPU (Apple M4 Max), with single-threaded numerical execution requested through the standard threading-environment variables; they characterize the computational footprint of the algorithms, not the latency of an IoT node. The AF-BKM model serializes to 5.5 KB at $d = 37$, 6.1 KB at $d = 39$ and 52.1 KB at $d = 115$ in closed-form fp32 (mean, inverse covariance, two scalars); this is the deployment payload, not the live memory footprint of the Python implementation. Fitting on 2000 packets takes 4–38 ms, and classifying a packet takes 11–27 µs on the benchmark CPU, while the per-round uplink is 160–472 B. The two timing columns measure different regimes and are not expected to agree by division: the batch column is a single vectorized call over a 20,000-packet array, so it reports throughput, whereas the per-packet column times individual single-row calls in a loop and is therefore dominated by fixed per-call Python/NumPy dispatch overhead. The loop uses 2000 repetitions for AF-BKM and 200 for each comparator, and no warm-up iteration precedes the timed batch call, so the columns are indicative of relative cost rather than a uniformly controlled benchmark. The memory column reports peak Python-traced allocation (tracemalloc), which covers allocations made through the Python allocator only and is not a measure of total process memory; it is measured in a separate pass, so it cannot distort the timing numbers. Against the comparators, the gap is architectural, with the caveat that their sizes are serialized Python objects, an implementation-level rather than platform-independent measure: instance-based detectors must store training samples, so their serialized models are 11 to 253 times larger (63 KB for OCSVM up to 1.4 MB for LOF, against 5.5 KB), and their scoring cost grows with the stored set. We deliberately report no energy figures: multiplying desktop latency by the nominal power of a different device

class would not be a valid energy measurement, and a physical study on target hardware is left to future work. As context for deployability, the worker-side inference of AF-BKM is the same single quadratic form as the original BKM, which was implemented and timed on an Arduino Nano 33 microcontroller in the prior work [9]; AF-BKM changes only the coordinator update, which never runs on the device. We did not conduct a new hardware deployment in this study.

Table 13. AF-BKM worker-side resource profile on the three datasets (benchmark CPU: Apple M4 Max; mean ± std over 3 seeds). Timings are desktop benchmarks, model and uplink sizes are closed-form fp32 values, and peak allocation is Python-traced rather than total process memory.

Dataset	d	Model (KB)	Uplink (B/Round)	Fit (ms)	Score 20k (ms)	Per-Packet (us)	Peak Traced Fit (KB)
NSL-KDD	37	5.5	160	3.9 ± 0.1	34.9 ± 0.6	11.2 ± 0.1	1452 ± 0
UNSW-NB15	39	6.1	168	4.2 ± 0.0	38.6 ± 0.3	11.4 ± 0.0	1517 ± 0
N-BaIoT	115	52.1	472	38.4 ± 0.1	374.1 ± 1.1	26.6 ± 0.4	4075 ± 0

Table 14. Exploratory desktop resource comparison on NSL-KDD (benchmark CPU: Apple M4 Max; one seeded split). AF-BKM model size is a closed-form fp32 serialization; comparator sizes are serialized Python objects. Peak allocation is Python-traced rather than total process memory.

Detector	Model (KB)	Fit (ms)	Score 20 k (ms)	Per-Packet (us)	Peak Traced Score (KB)
Bkmeans-Mahalanobis	5.5	3.9	34.3	11.1	11,628
OCSVM	63	9.1	116.7	43.6	6095
IsolationForest	1005	67.0	55.5	3553.0	9690
LOF	1391	33.2	194.6	132.5	37,347
KDE	633	148.3	1448.8	121.2	5942
KNN	579	9.7	87.6	87.8	7523

7. Discussion, Limitations, and Future Work

The longitudinal results and ablations identify the same failure mechanism: the merge-induced precision decay comes from the nearest-anomaly threshold tightening (Equation (4)), and replacing it with a benignly anchored recalibration (Equation (8)) removes the decay on all three datasets without abandoning the statistics-only, semi-supervised, resource-conscious IoT edge design. AF-BKM trades a controllable amount of recall for substantially higher and stable precision and a 30–64% lower mean FPR; in security operations, where alert fatigue is costly, this is usually the desired trade-off. Those figures are measured under the MAD default, which holds the false-positive level steady across merges without encoding a chosen target; the operating point can be made explicit by switching r to the target-FPR rule.

7.1. Position Relative to Recent Federated IDS

The differential analysis of Section 3 can now be read against shared evidence, since FedMSE [14] also evaluates on N-BaIoT. The two systems answer different questions under different protocols: FedMSE trains shrink autoencoders per gateway, aggregates *weights* with an MSE-aware rule, splits N-BaIoT at the device level, and reports accuracy near 0.97; AF-BKM shares only a d+3-value statistics summary (472 bytes per round at $d = 115$), runs a windowed non-IID stream, and reaches an F1 of 0.91–0.96 with full recall at a stable false-positive level (0.106 under the MAD default; operator-selectable when $r = \tau_{\mathrm{FPR}}$). The numbers are not directly comparable, and we do not rank the systems; the relevant contrast is architectural and procedural. An autoencoder weight update carries the full parameter tensor, whereas AF-BKM's upload is fixed at d+3 values, regardless of any model capacity, and the systems in our comparison do not study merge-time threshold

recalibration, which is the specific failure AF-BKM repairs. Reconstruction attacks put the same contrast in privacy terms: shared gradients can be inverted to recover training inputs [33], and model inversion can recover class-representative inputs from a trained model's outputs and confidence values [34]. AF-BKM transmits windowed feature means and a few scalars, which reduces the exposed object relative to gradients but is not a privacy guarantee: aggregate means can still leak information, particularly for small or highly structured windows, and a formal inference analysis with differentially private noising of the reported means remains for future work. Backdoor attacks in their classical form implant a trigger pattern into a trainable model [35]; a centroid-plus-threshold detector offers no neural trigger-memorization pathway, but a poisoned centroid or threshold can still carve out an attacker-favored acceptance region. The mean-shift faulty update and the anchor-contamination scenario of Sections 6.6 and 6.7 measure two such surfaces; they do not constitute complete backdoor coverage.

7.2. Position Relative to the Previous BKM Line

The published cross-layer federated BKM system showed that compact statistic exchange can improve collaborative recall in IoT IDS [9]; the ensemble extension then showed that false positives can be suppressed by adding complementary novelty detectors around BKM [10]. AF-BKM is intentionally different from both. It does not add model capacity at the edge, and it does not require a downstream vote. Instead, it fixes the coordinator behavior that made the original federated detector lose precision after merges. In this sense, AF-BKM can be used either as a simpler alternative when only a statistics-only detector is desired or as a cleaner base layer for a future ensemble whose second-stage models no longer need to compensate for a miscalibrated global threshold.

7.3. Limitations

The main limitation is *anchor trust*: the false-positive level is held constant with respect to the trusted benign anchor, so the property is anchor-relative rather than local when a worker's benign distribution drifts far from it, and under the MAD default, it is a stable level rather than an operator-chosen one, and it further assumes secure coordinator storage of the anchor. Section 6.7 quantifies an imperfect anchor: the FPR stays controlled at every tested contamination level, while recall is dataset-dependent, roughly stable through 2% on NSL-KDD but already declining on UNSW-NB15 and dropping sharply on both from 5%; the loss is driven by the non-robust mean and covariance rather than the threshold rule. The non-IID setting is also still incomplete. This paper models heterogeneity primarily through label skew with a small viability floor; feature-space skew, per-device behavioral drift, and changing traffic mixtures deserve deeper treatment. On the hardware side, we report desktop CPU timing, Python-traced peak allocation and closed-form fp32 serialization sizes; on-device feasibility of the unchanged inference kernel is supported contextually by the Arduino implementation of the prior work [9], but this study includes no new physical edge deployment and reports no energy measurements because no target hardware or power instrumentation was available. Although N-BaIoT consists of traffic captured from real commercial devices under real infections, all three corpora are recorded datasets; a live pilot on an operational network, with the privacy and ethics apparatus it requires, is beyond the scope of this study. The UNSW-NB15 results should be read carefully: AF-BKM exposes a useful precision/FPR-oriented operating point, but the lower recall at stricter targets means that recall-critical deployments may need a looser target or an additional detector, and the evasion analysis of Section 6.8 shows that the recall-halving budget is smallest and most seed-dependent on this narrow-margin traffic. Finally, the robustness study covers the selected faulty-worker updates, anchor contamination and white-box

evasion; it establishes no worst-case guarantee and does not cover adaptive adversaries that jointly optimize against the trust gate, the blend and the anchor [32].

7.4. Future Work

The most immediate extension is streaming concept drift, using lightweight drift detectors or adaptive anchor refresh policies so that benign calibration remains valid over time [21]. The anchor-contamination results argue separately for robust estimation underneath the robust threshold: replacing the sample mean and covariance with a minimum-covariance-determinant estimator [36] would harden the calibration against dirty commissioning windows, since shrinkage [37] improves conditioning but does not, by itself, confer outlier robustness and diagonal, block-diagonal or low-rank covariance summaries could additionally cut communication. A physical deployment pass also remains to be done: on-device timing, memory and measured (rather than estimated) energy on microcontroller hardware, ideally together with a live pilot. Evaluation on further device-centric corpora such as Bot-IoT is a separate open item. Over a longer horizon, AF-BKM could be combined with the earlier ensemble idea, providing a stable, calibrated first-stage BKM signal while lightweight local detectors are invoked only near ambiguous decision regions, which the evasion analysis suggests is where they are most needed.

8. Conclusions

We revisited a published federated, semi-supervised IoT intrusion detector whose periodic merges, while raising recall, systematically eroded precision and inflated the false-positive rate and whose threshold was hand-tuned. AF-BKM repairs the federated mechanism with two label-free, statistics-only changes: an adaptive threshold that requires no manual percentile search and a robust, benignly anchored aggregation that recalibrates the threshold to a stable, anchor-referenced false-positive level instead of tightening it toward the nearest anomaly. With MinMax scaling fit only on the benign baseline and a disclosed label-blind, full-corpus, near-constant filter, AF-BKM removes the measured merge-induced precision decay in the evaluated protocol and reduces the mean false-positive rate by 30–64% depending on the dataset, with statistically significant improvements across 10 seeds on NSL-KDD, UNSW-NB15 and the real-device N-BaIoT corpus; it preserves recall on NSL-KDD and N-BaIoT and exposes an explicit, operator-controlled precision–recall trade-off on UNSW-NB15 while retaining a Mahalanobis core that serializes to 5.5–52 KB and uploads a $d+3$-value summary (160–472 B) per merge round, 94.6–98.3% less than the same summary with a covariance upper triangle. The stress tests are consistent with the design: the threshold is recalibrated on the commissioning anchor rather than on worker-reported values, so fabricated candidates have no direct path to it, though a fabricated mean still reaches it indirectly through the blended centroid; the anchor-referenced false-positive level remains stable under percent-level anchor contamination while recall sensitivity is dataset-dependent; and the cost of white-box mimicry tracks each dataset's benign–attack margin. Future work will add streaming drift adaptation, robust covariance and vector-update defenses, and a physical on-device deployment with measured energy.

Author Contributions: Conceptualization, M.A.S. and J.A.; methodology, M.A.S. and J.A.; software, J.A.; validation, M.A.S. and J.A.; writing—original draft, M.A.S. and J.A.; writing—review and editing, M.A.S. and J.A.; supervision, J.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: NSL-KDD and UNSW-NB15 are publicly available. The N-BaIoT corpus is used through the prepared 50-client non-IID partition distributed with the FedMSE code release (commit `d126885`); the exact preparation and validation steps are documented in the repository README. The AF-BKM source code, the experiment scripts, and the generators for the empirical result figures and tables are publicly available at https://github.com/josephazar/af-bkm (accessed on 13 August 2026).

Acknowledgments: During the preparation of this manuscript, the authors used Anthropic's Claude (Claude Opus 5) for language editing and drafting assistance in the writing and documentation of the text. The research question, experimental design, choice of methods, analysis and interpretation of the results are the authors' own. The authors reviewed, verified and edited all generated content, independently checked all reported numerical results against the released code, and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Campos, E.M.; Saura, P.F.; González-Vidal, A.; Hernández-Ramos, J.L.; Bernabe, J.B.; Baldini, G.; Skarmeta, A. Evaluating Federated Learning for intrusion detection in Internet of Things: Review and challenges. *Comput. Netw.* **2022**, *203*, 108661. [CrossRef]
2. Rahman, S.A.; Tout, H.; Talhi, C.; Mourad, A. Internet of Things intrusion detection: Centralized, on-device, or federated learning? *IEEE Netw.* **2020**, *34*, 310–317. [CrossRef]
3. McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; Arcas, B.A. Communication-efficient learning of deep networks from decentralized data. In *Artificial Intelligence and Statistics*; PMLR: Fort Lauderdale, FL, USA, 2017; pp. 1273–1282.
4. Belenguer, A.; Navaridas, J.; Pascual, J.A. A review of federated learning in intrusion detection systems for IoT. *arXiv* **2022**, arXiv:2204.12443.
5. Khraisat, A.; Alazab, A.; Singh, S.; Jan, T.; Gomez, A., Jr. Survey on federated learning for intrusion detection system: Concept, architectures, aggregation strategies, challenges, and future directions. *ACM Comput. Surv.* **2024**, *57*, 1–38. [CrossRef]
6. Al Saleh, M.; Shbaro, R.; Azar, J. A novel IoT security framework combining X25519 with NIST lightweight Ascon encryption and hybrid transform-domain steganography. *Telecom* **2026**, *7*, 40. [CrossRef]
7. Blanchard, G.; Lee, G.; Scott, C. Semi-supervised novelty detection. *J. Mach. Learn. Res.* **2010**, *11*, 2973–3009.
8. Hajj, S.; El Sibai, R.; Barada, A.; Abdo, J.B.; Demerjian, J.; Guyeux, C.; Makhoul, A.; Ginhac, D. Cluster-based Sampling Algorithm for Lightweight IoT Intrusion Detection System. In *Proceedings of the International Conference on Security and Management (SAM'22), Part of CSCE'22*, Las Vegas, NV, USA, 25–28 July 2022.
9. Hajj, S.; Azar, J.; Bou Abdo, J.; Demerjian, J.; Guyeux, C.; Makhoul, A.; Ginhac, D. Cross-layer federated learning for lightweight IoT intrusion detection systems. *Sensors* **2023**, *23*, 7038. [CrossRef] [PubMed]
10. Hajj, S.; Azar, J.; Bou Abdo, J.; Demerjian, J.; Makhoul, A.; Ginhac, D. Cross-layer federated heterogeneous ensemble learning for lightweight IoT intrusion detection system. In *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics (DSAA)*; IEEE: New York, NY, USA, 2023; pp. 1–10. [CrossRef]
11. Belarbi, O.; Spyridopoulos, T.; Anthi, E.; Mavromatis, I.; Carnelli, P.; Khan, A. Federated deep learning for intrusion detection in IoT networks. In *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*; IEEE: New York, NY, USA, 2023; pp. 237–242.
12. Sáez-de-Cámara, X.; Flores, J.L.; Arellano, C.; Urbieta, A.; Zurutuza, U. Clustered federated learning architecture for network anomaly detection in large scale heterogeneous IoT networks. *Comput. Secur.* **2023**, *131*, 103299. [CrossRef]
13. Shrivastava, A.; Vamsi, P.R. Mahalanobis Distance-Based Supervised and Semi-Supervised Machine Learning Methods for Anomaly Detection in IoT Sensor Data. *Int. J. Sens. Wirel. Commun. Control* **2026**, *16*. [CrossRef]
14. Nguyen, V.T.; Beuran, R. FedMSE: Semi-supervised federated learning approach for IoT network intrusion detection. *Comput. Secur.* **2025**, *151*, 104337. [CrossRef]
15. Attota, D.C.; Mothukuri, V.; Parizi, R.M.; Pouriyeh, S. An ensemble multi-view federated learning intrusion detection for IoT. *IEEE Access* **2021**, *9*, 117734–117745. [CrossRef]
16. Bouzinis, P.S.; Radoglou-Grammatikis, P.; Makris, I.; Lagkas, T.; Argyriou, V.; Papadopoulos, G.T.; Sarigiannidis, P.; Karagiannidis, G.K. StatAvg: Mitigating data heterogeneity in federated learning for intrusion detection systems. *IEEE Trans. Netw. Serv. Manag.* **2025**, *22*, 2944–2955. [CrossRef]

17. Nakıp, M.; Gül, B.C.; Gelenbe, E. Decentralized online federated G-network learning for lightweight intrusion detection. In *Proceedings of the IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*; IEEE: New York, NY, USA, 2023; pp. 1–8.
18. Lima, M.F.; Zarpelão, B.B.; Sampaio, L.D.H.; Rodrigues, J.J.; Abrao, T.; Proença, M.L. Anomaly detection using baseline and K-means clustering. In *Softcom 2010, 18th International Conference on Software, Telecommunications and Computer Networks*; IEEE: New York, NY, USA, 2010.
19. Melnykov, I.; Melnykov, V. On *K*-means algorithm with the use of Mahalanobis distances. *Stat. Probab. Lett.* **2014**, *84*, 88–95. [CrossRef]
20. Wu, Y.; Liu, L.; Yu, Y.; Chen, G.; Hu, J. Online ensemble learning-based anomaly detection for IoT systems. *Appl. Soft Comput.* **2025**, *173*, 112931. [CrossRef]
21. Gama, J.; Žliobaitė, I.; Bifet, A.; Pechenizkiy, M.; Bouchachia, A. A survey on concept drift adaptation. *ACM Comput. Surv.* **2014**, *46*, 44. [CrossRef] [PubMed]
22. Azar, J.; Al Saleh, M.; Couturier, R.; Noura, H. Text mining and unsupervised deep learning for intrusion detection in smart-grid communication networks. *IoT* **2025**, *6*, 22. [CrossRef]
23. Yassine, H.A.K.; El Saleh, M.; Nakhal, B.E.; El Chakik, A. Centralized two-tiered tree-based intrusion-detection system (C2T-IDS). *IoT* **2025**, *6*, 67. [CrossRef]
24. Otsu, N. A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.* **1979**, *9*, 62–66. [CrossRef]
25. Tavallaee, M.; Bagheri, E.; Lu, W.; Ghorbani, A.A. A detailed analysis of the KDD CUP 99 data set. In *Proceedings of the IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*; IEEE: New York, NY, USA, 2009; pp. 1–6. [CrossRef]
26. Moustafa, N.; Slay, J. UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In *Proceedings of the Military Communications and Information Systems Conference (MilCIS)*; IEEE: New York, NY, USA, 2015; pp. 1–6. [CrossRef]
27. Meidan, Y.; Bohadana, M.; Mathov, Y.; Mirsky, Y.; Shabtai, A.; Breitenbacher, D.; Elovici, Y. N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Comput.* **2018**, *17*, 12–22. [CrossRef]
28. Al Saleh, M.; Finance, B.; Taher, Y.; Jaber, A.; Luff, R. Online classification of high gamma dose rate incidents. *Eng. Proc.* **2022**, *18*, 28. [CrossRef]
29. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
30. Blanchard, P.; El Mhamdi, E.M.; Guerraoui, R.; Stainer, J. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Advances in Neural Information Processing Systems*; Curran Associates, Inc.: Red Hook, NY, USA, 2017; pp. 119–129.
31. Yin, D.; Chen, Y.; Kannan, R.; Bartlett, P. Byzantine-robust distributed learning: Towards optimal statistical rates. In *International Conference on Machine Learning*; PMLR: Fort Lauderdale, FL, USA, 2018; pp. 5650–5659.
32. Fang, M.; Cao, X.; Jia, J.; Gong, N.Z. Local model poisoning attacks to Byzantine-robust federated learning. In Proceedings of the 29th USENIX Security Symposium (USENIX Security 20), Boston, MA, USA, 12–14 August 2020; pp. 1605–1622.
33. Zhu, L.; Liu, Z.; Han, S. Deep leakage from gradients. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*; Curran Associates, Inc.: Red Hook, NY, USA, 2019; pp. 14774–14784.
34. Fredrikson, M.; Jha, S.; Ristenpart, T. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*; Association for Computing Machinery: New York, NY, USA, 2015; pp. 1322–1333.
35. Gu, T.; Liu, K.; Dolan-Gavitt, B.; Garg, S. BadNets: Evaluating backdooring attacks on deep neural networks. *IEEE Access* **2019**, *7*, 47230–47244. [CrossRef]
36. Rousseeuw, P.J.; Van Driessen, K. A fast algorithm for the minimum covariance determinant estimator. *Technometrics* **1999**, *41*, 212–223. [CrossRef]
37. Ledoit, O.; Wolf, M. A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivar. Anal.* **2004**, *88*, 365–411. [CrossRef]

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